

BitFlip

CloudSherpa

User Manual

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1 | Welcome to CloudSherpa

CloudSherpa is an intelligent cloud cost management platform designed to help organizations gain complete visibility into their cloud spending across multiple cloud providers. By combining cloud usage, costs and insights into a single, intuitive finOps dashboard, it eliminates the need to navigate between multiple cloud provider platforms.

The platform enables users to monitor cloud expenditure, identify unexpected cost increases and receive proactive recommendations to optimize resource usage and reduce unnecessary spending. Through real-time insights and predictive analytics, CloudSherpa helps organizations make informed financial decisions while maintaining efficient cloud operations.

By providing a unified view of multi-cloud environments and actionable cost optimization recommendations, CloudSherpa empowers businesses to control cloud expenses, maximise the values of their cloud investments and confidently manage their cloud platform.

Do I need to be a technical expert?

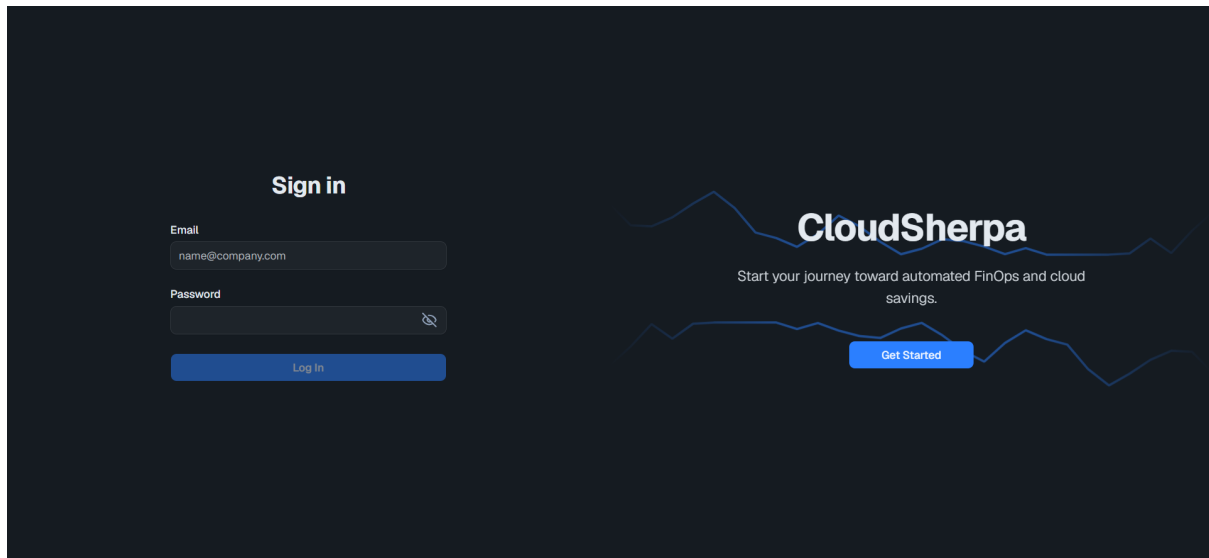
CloudSherpa is designed for both technical and non-technical users. It features a user-friendly dashboard that presents insights in a clear, intuitive manner, which enables users of any technical level to easily understand what is occurring.

2 | Getting Started

2.1 Accessing the System and Logging In

How to access the live deployment of CloudSherpa:

1. Navigate to <https://cloudsherpa.gjjcs.org/>



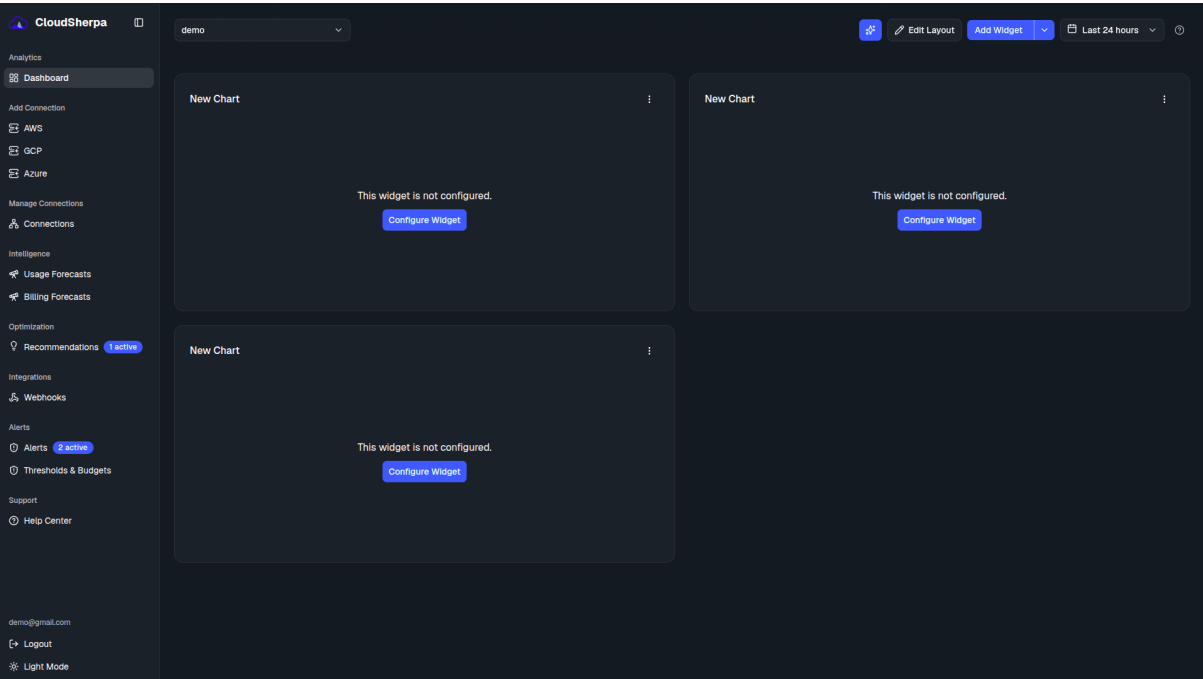
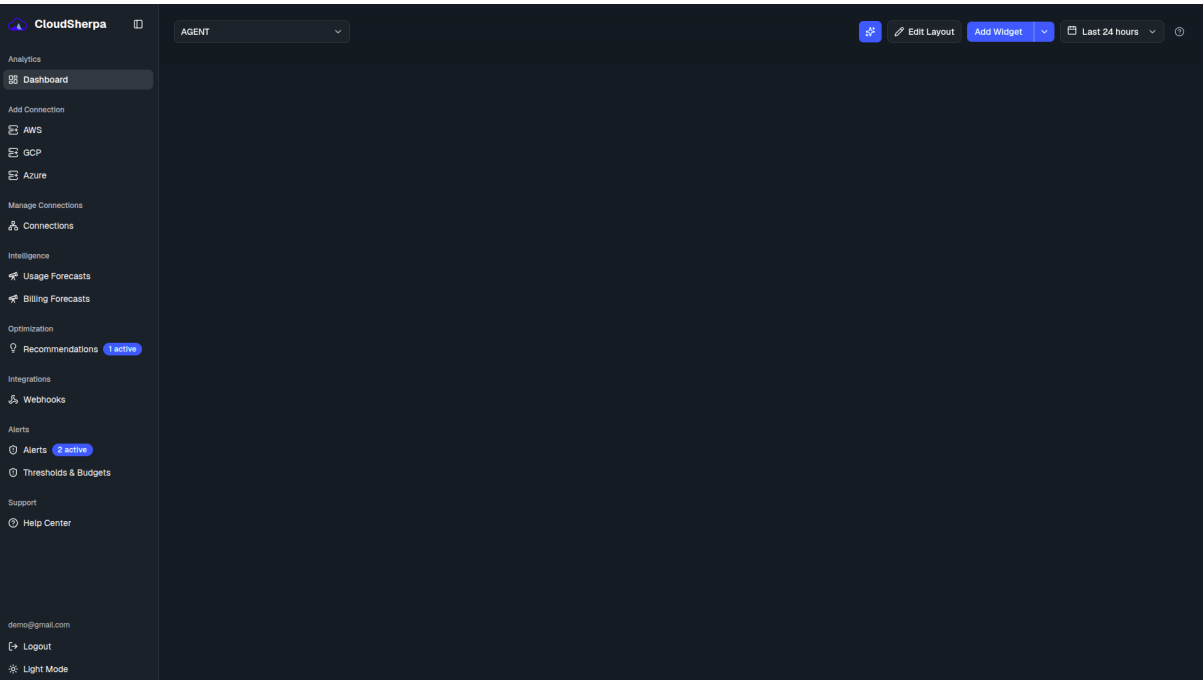
To create an account with CloudSherpa, follow these steps:

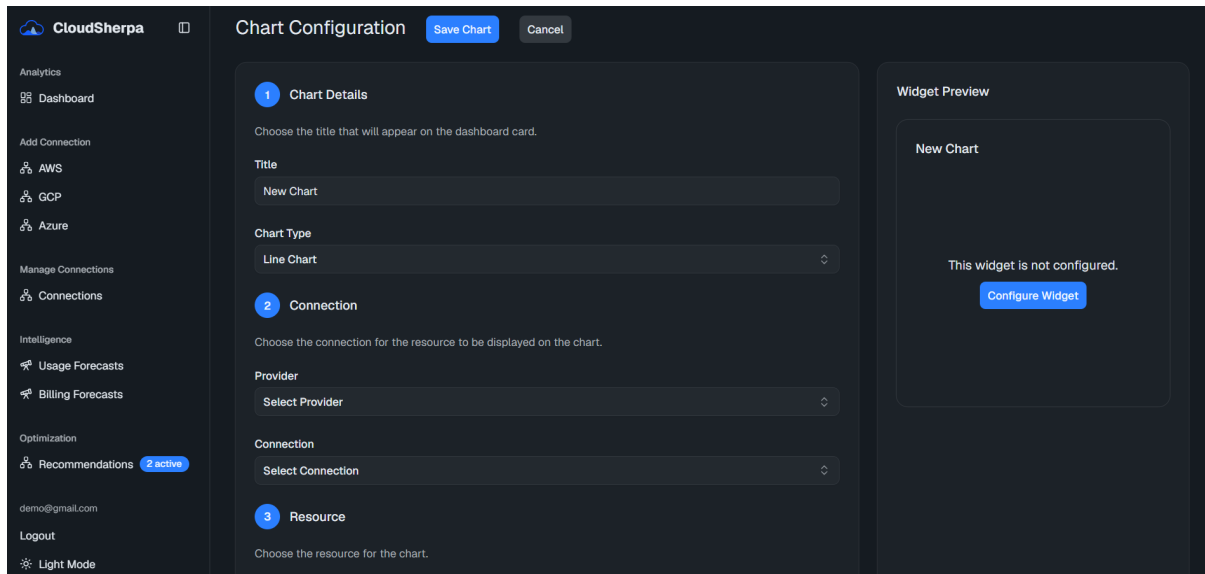
1. If you are a new user:
 - (a) Click the **Register** button.
 - (b) Fill in the required information.
 - (c) Create a secure password and confirm your registration.
2. If you already have an account:
 - (a) Click the **Login** button.
 - (b) Enter your registered email and password.
 - (c) Click **Login** to access your dashboard.

By following these steps, you can immediately begin using CloudSherpas features.

2.2 The Main Dashboard

Upon logging in the user is directed to the dashboard:





Key areas of the dashboard:

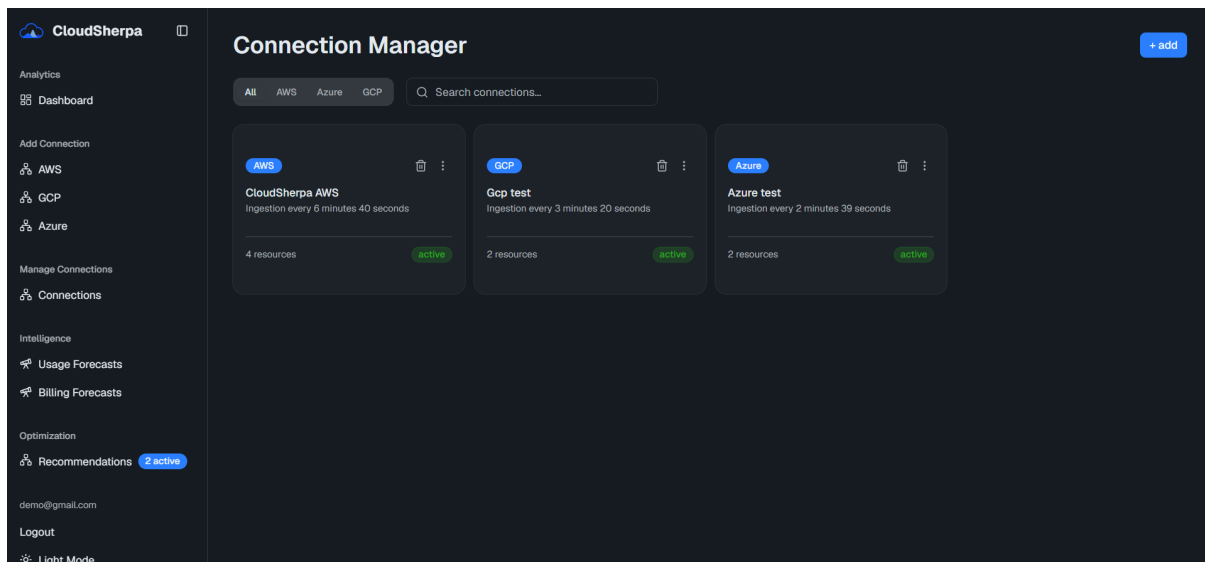
- **Navigation bar:** Users are able to navigate to various connections such as AWS. They are able to log out via the navigation bar or go back to the dashboard. The navigation bar is collapsible to give users a better view of what is being displayed on their screen.
- **Period selector:** The user can select which time period they would like to view the data being displayed. There are fixed period selectors such as 'last 7 days' or they can create a custom period that they desire.
- **Multiple dashboard:** More than one dashboard can be created via the dropdown. The users can also navigate to dashboards that were previously created via the dropdown.
- **Customize widgets:** Users can design the dashboard in a way they desire. The widgets size can be adjusted and their positions can be changed. This can be done by using the pen icon (next to the dropdown). They can further choose a graph they would like the data to be displayed in, from the options provided. The name of the widget can be customized and different types of data can be selected to be displayed eg.CPU.

3 | Core Features and Usage

These are the five core features of CloudSherpa

3.1 Feature 1: Connection Manager

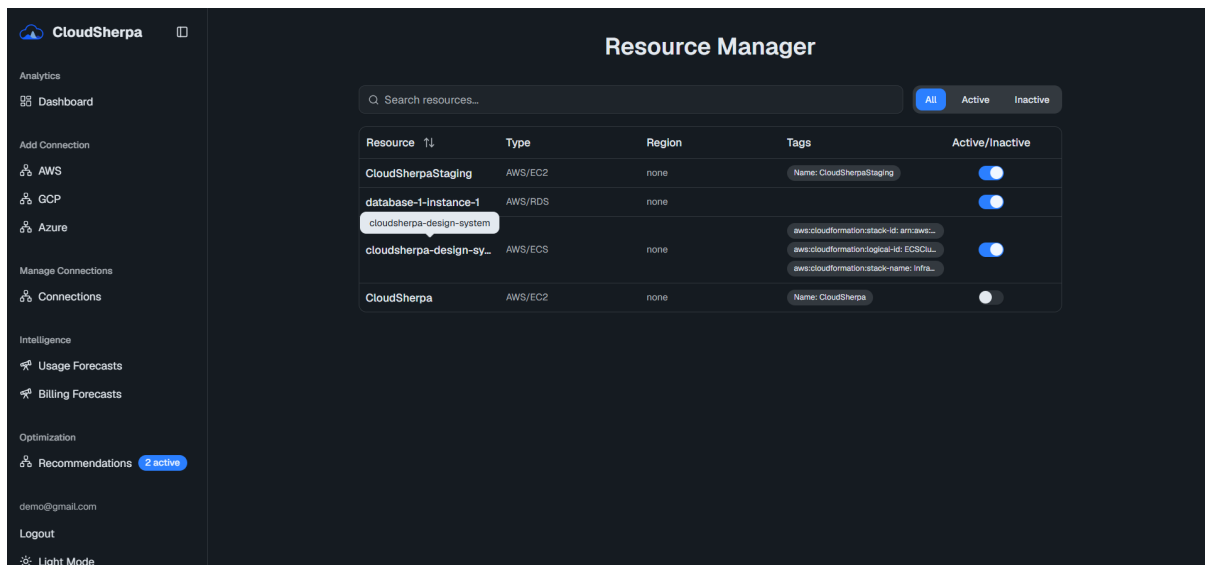
The connection manager is utilized to add multiple AWS connections where their services can be monitored as well as specific resources.



1. Locating connection manager - Navigate to the Connection manager (under Manage Connections) in the side navigation bar
2. For adding a connection you can navigate to the connection manager and click on the add button or you can use the side navigation bar to choose which connection you would like to establish.
3. Managing your connections - by clicking on the view details option. (will get by clicking on the 3 dots) You can change the name of the connections and view the resources that is associated with that connection. Multiple connections can be added and deleted.

3.2 Feature 2: Resource Manager

The resource manager is used to manage your connections. You can make certain resources active and inactive. You can filter and search for a resource.



1. Locating the resource manager - In the connection manager configuration (was shown in previous feature), click on the edit resources button. You can successfully configure your resources here.
2. Active/Inactive resources - toggle the resources to determine which resource should be monitored for that particular service or not.
3. Search and filter resources - Use the search bar to search for a resource or to filter click on the Resource name.
4. tooltips have also been included for resource names, types and tags hence the full view can be viewed when hovered over them.

3.3 Feature 3: Scheduler

The scheduler scans the database every 30 seconds for the usage scheduler and 5 minutes for the billing scheduler.

We read in a field that informs us that an account is due and if they are they get enqueued to Jobrunr background jobs queue which is managed by Jobrunr.

Jobrunr spins threads in the background, up to 8 in our configuration. Each job attempts ingestion for only one account. If it fails, it retries 3 times and if it has still failed

the appropriate field in the database will not get updated which mean that when the scheduler scans again that account will be scheduled again.

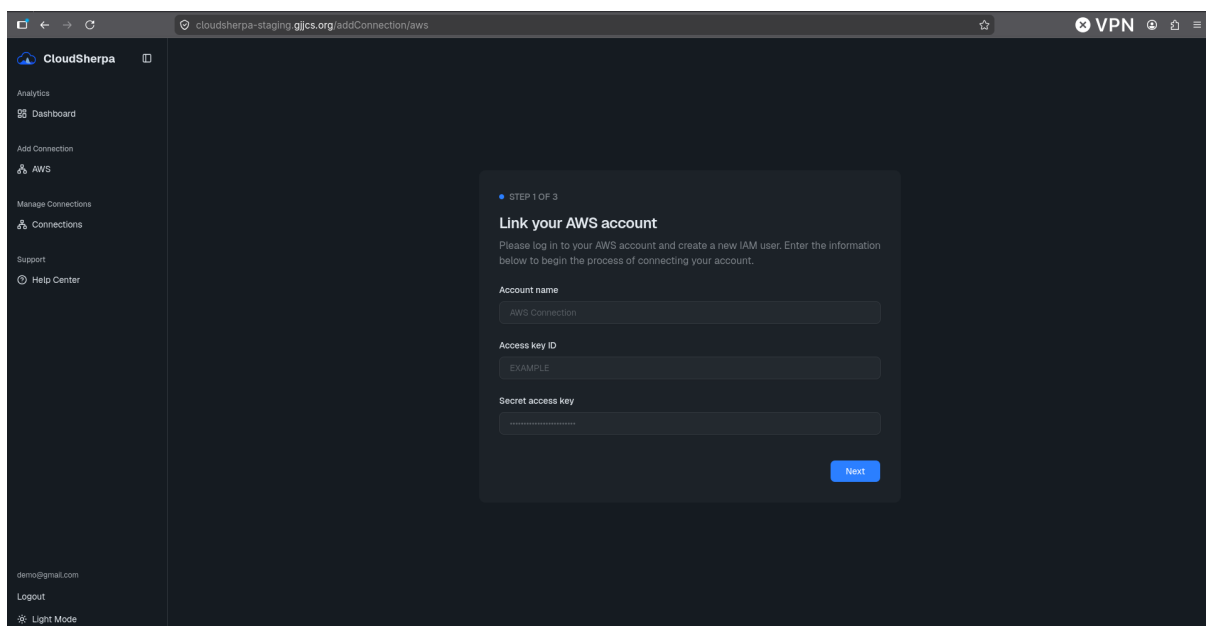
If it was successful, the appropriate next ingestion field gets updated in the database such that the next scan will not pick the same job again

3.4 Feature 4: Setup wizard

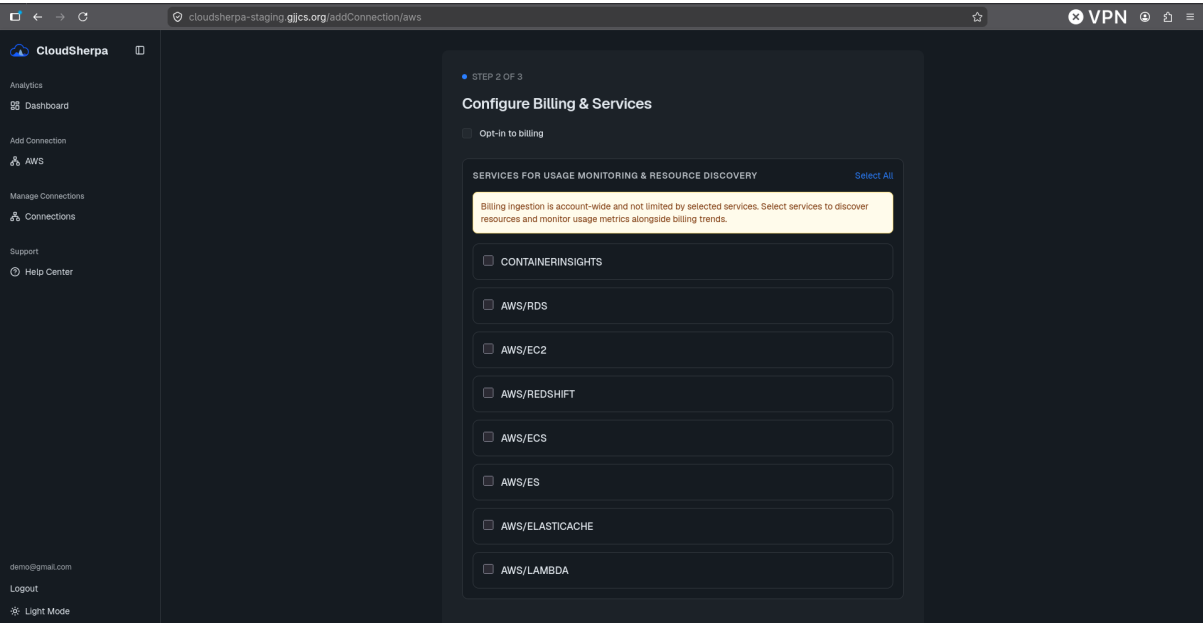
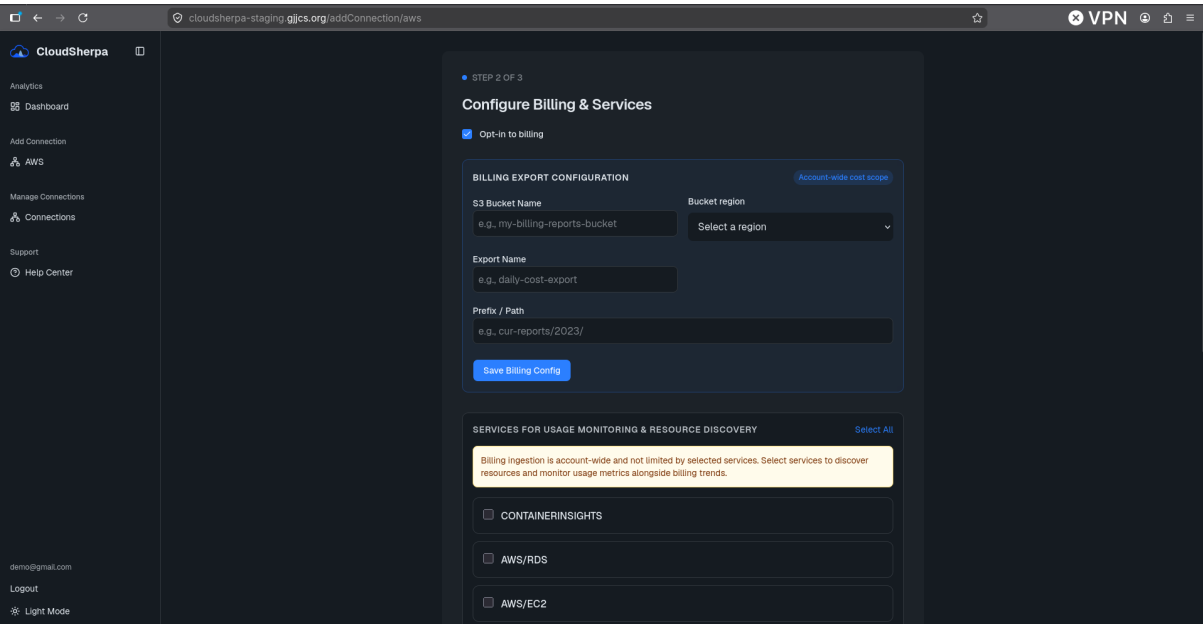
CloudSherpa provides three setup wizards for connecting to different cloud providers:

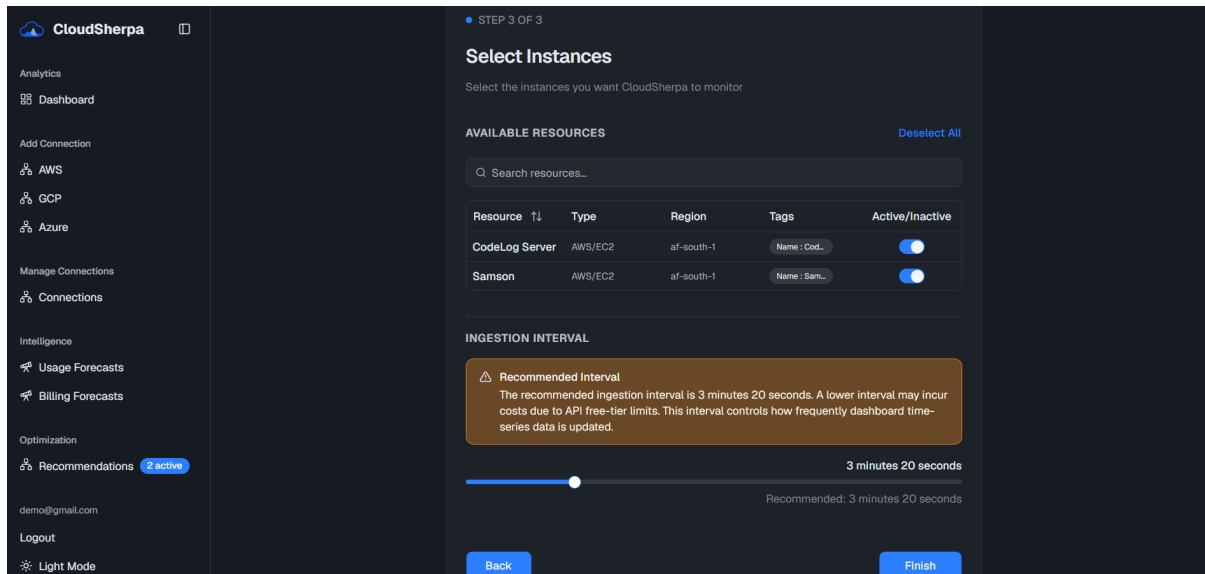
a) AWS Setup Wizard

The setup wizard consists of 3 steps, which each user has to go through each time they want to establish an AWS connection.



The screenshot shows the CloudSherpa web application interface. On the left is a dark sidebar with navigation links: Analytics, Dashboard, Add Connection, AWS, Manage Connections, Connections, Support, and Help Center. At the bottom of the sidebar, it shows 'demo@gmail.com', 'Logout', and 'Light Mode'. The main content area is dark and displays a 'STEP 1 OF 3' wizard titled 'Link your AWS account'. Below the title, it says 'Please log in to your AWS account and create a new IAM user. Enter the information below to begin the process of connecting your account.' There are three input fields: 'Account name' (with 'AWS Connection' as a placeholder), 'Access key ID' (with 'EXAMPLE' as a placeholder), and 'Secret access key' (with a masked placeholder). A blue 'Next' button is at the bottom right of the form.





1. Step 1 - Provide the credentials for IAM user. You can choose what you would like to call the connection (can be changed at any time under Connections).
2. Step 2 - Select the services which you would like CloudSherpa to monitor. The billing in this step is optional and you can also opt in or out of usage monitoring for specific services.
3. Step 3 - Displays the resources that are associated with the services that were in the previous step. You can select which ones you would like CloudSherpa to monitor (can be managed anytime under Connections).

b) GCP Setup Wizard

The setup wizard consists of 3 steps, which each user has to go through each time they want to establish a GCP connection.

CloudSherpa

Analytics

Dashboard

Add Connection

AWS

GCP

Azure

Manage Connections

Connections

Intelligence

Usage Forecasts

Billing Forecasts

Optimization

Recommendations 2 active

demo@gmail.com

Logout

Light Mode

STEP 1 OF 3

Link your GCP account

Please download the service account key as a JSON file and upload it below to begin connecting your account

Account name

Connection name

Service account key

Drag and drop your file here

or

Click to browse files

Next

CloudSherpa

Analytics

Dashboard

Add Connection

AWS

GCP

Azure

Manage Connections

Connections

Intelligence

Usage Forecasts

Billing Forecasts

Optimization

Recommendations 2 active

demo@gmail.com

Logout

Light Mode

STEP 2 OF 3

Select service

Choose which GCP service you want to monitor.

Opt-in to billing

BILLING EXPORT CONFIGURATION

Account-wide cost scope

Cloud Billing Account ID

Enter the Google Cloud Billing account ID for this connection.

XXXXXXXX-XXXXXX-XXXXXX

Billing Export Dataset ID

Enter the BigQuery dataset that contains the detailed billing export tables.

example_dataset

SERVICES WE OFFER

Deselect All

GCP/ComputeEngine

PERMISSIONS

The image displays two screenshots of the CloudSherpa web application interface, showing the setup process for monitoring GCP services.

Top Screenshot: STEP 2 OF 3 - Select service

The interface shows the 'Select service' step. The user is prompted to 'Choose which GCP service you want to monitor.' There is an 'Opt-in to billing' checkbox. Under 'SERVICES WE OFFER', 'GCP/ComputeEngine' is selected. Under 'PERMISSIONS', the following permissions are listed: Compute Viewer, Service Usage Viewer, Cloud Asset Viewer, and Monitoring Viewer. 'Back' and 'Next' buttons are at the bottom.

Bottom Screenshot: STEP 3 OF 3 - Select instances

The interface shows the 'Select instances' step. The user is prompted to 'Select the instance you want CloudSherpa to monitor.' Under 'AVAILABLE RESOURCES', there is a search bar and a table of resources:

Resource	Type	Region	Tags	Active/Inactive
cerberus-vpn~...	gce_instance	africa-south1...		☒
instance-2026...	gce_instance	africa-south1...	goog-ops-a...	☒

Below the table is the 'INGESTION INTERVAL' section, which includes a warning about the recommended interval (3 minutes 20 seconds) and a slider to adjust the interval. The slider is currently set to 3 minutes 20 seconds. 'Back' and 'Finish' buttons are at the bottom.

1. Step 1 - Enter an account name that you can refer to when looking for a particular connection. Along with that upload your service account key (it must be a JSON file).
2. Step 2 - The user has an option of selecting whether they would like to apply billing for this connection or not. They need to select all of the services they would like CloudSherpa to monitor. A list of permissions will appear that the user would need to grant for your newly created GCP IAM user.
3. Step 3 - Once all of the resources associated with your selected services have been discovered, you can choose which ones you would like to monitor. The ingestion interval slider is dependent on the resources that are chosen. The

interval can be adjusted but if a value selected is below the recommended value, the user will be warned of the consequences.

c) Azure Setup Wizard

The setup wizard consists of 3 steps, which each user has to go through each time they want to establish an Azure connection.

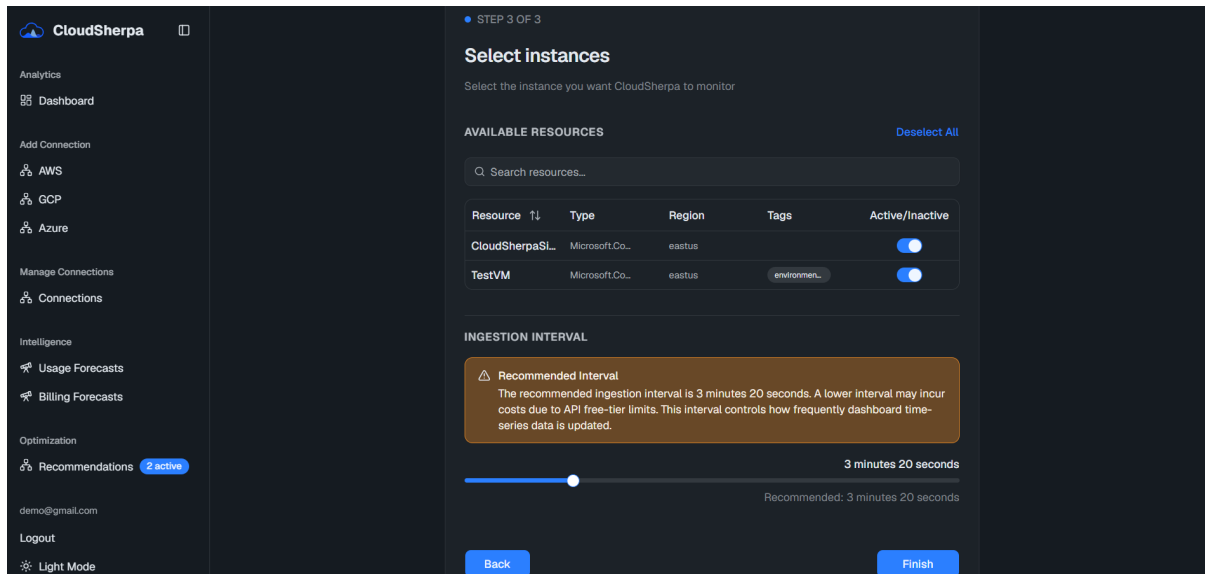
The image displays two screenshots of the CloudSherpa Azure Setup Wizard interface.

Top Screenshot (Step 1 of 3):

- Header:** CloudSherpa
- Left Sidebar:** Analytics, Dashboard, Add Connection (AWS, GCP, Azure), Manage Connections (Connections), Intelligence (Usage Forecasts, Billing Forecasts), Optimization (Recommendations - 2 active), demo@gmail.com, Logout, Light Mode.
- Main Content:**
 - STEP 1 OF 3**
 - Link your Azure account**
 - Account name:** Connection name
 - Subscription ID:** Azure subscription ID
 - Tenant ID:** Tenant ID
 - Client ID:** Client ID
 - Client Secret:** (password field)
 - Next** button

Bottom Screenshot (Step 2 of 3):

- Header:** CloudSherpa
- Left Sidebar:** Same as the top screenshot.
- Main Content:**
 - STEP 2 OF 3**
 - Select service**
 - SERVICES WE OFFER:** Deselect All, ☒ AZURE/VM
 - PERMISSIONS:**
 - Monitoring Reader
 - Reader
 - Back** and **Next** buttons

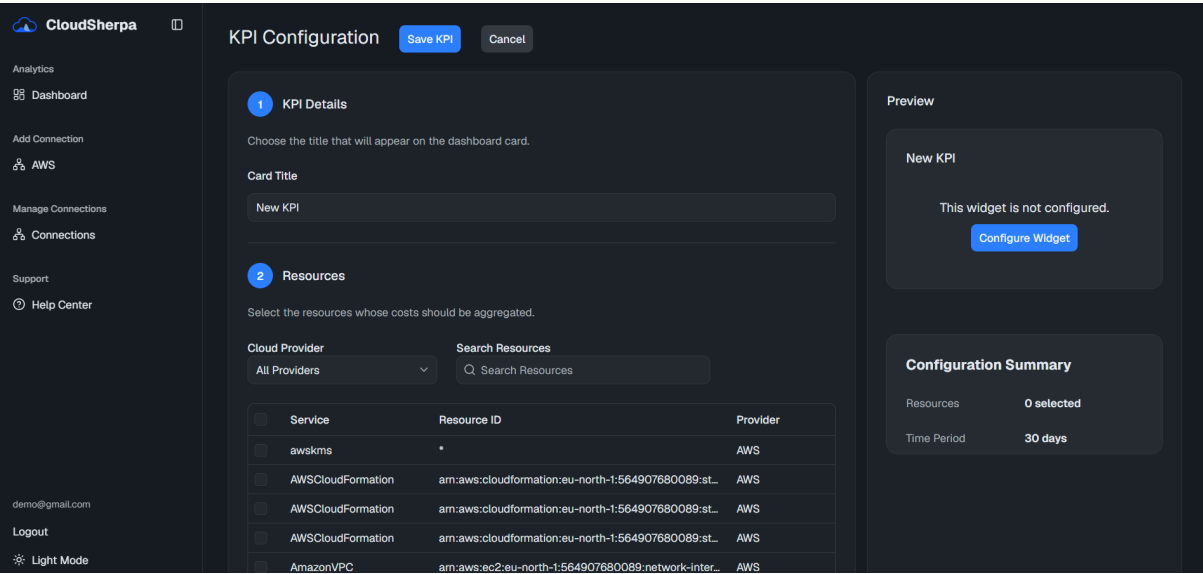
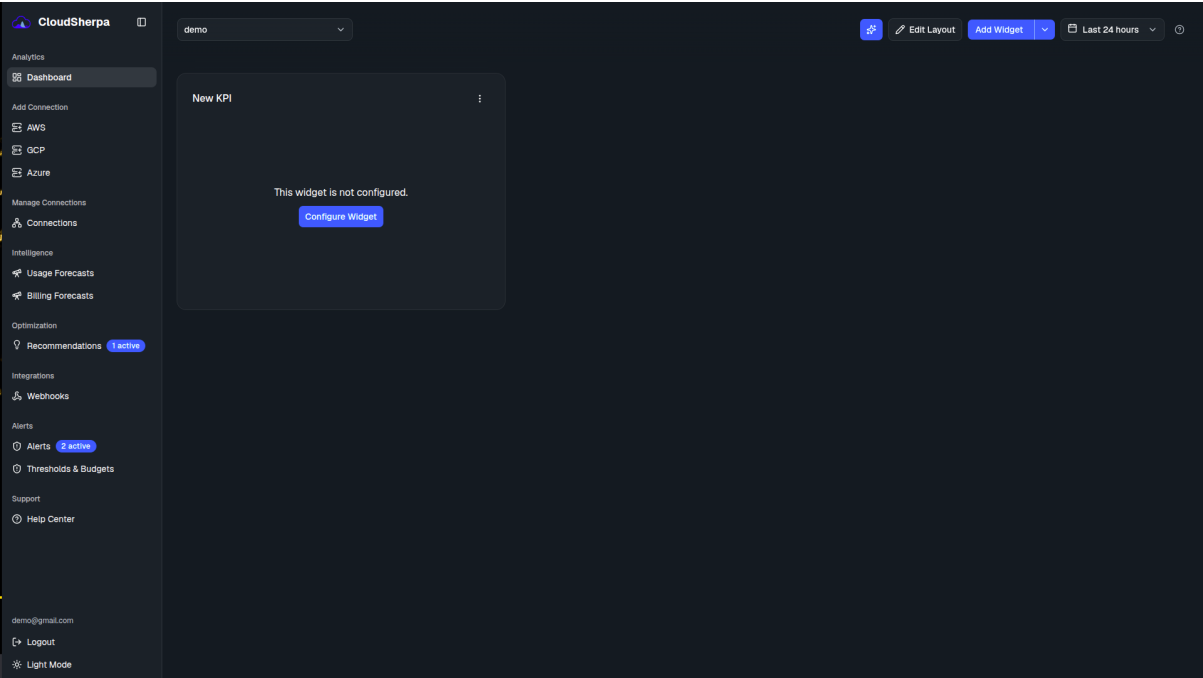


1. Step 1 - Add an account name which you can refer to when looking for a particular connection. Enter your Subscription ID, Tenant ID, Client ID and your Client Secret.
2. Step 2 - Select the services that would be monitored by CloudSherpa and you would then get a list of permissions that you would need to grant.
3. Step 3 - Once all of the resources associated with your selected services have been discovered, you can choose which ones you would like to monitor. The ingestion interval slider is dependent on the resources that are chosen. The interval can be adjusted but if a value selected is below the recommended value, the user will be warned of the consequences.

Some additional features that is available in all of the setup wizards include: Being able to make a certain resource active or inactive for CloudSherpa to monitor. The user can search for resources, Select all of the resources and sort the resources by using the tanstack table feature.

3.5 Feature 5: KPI widget, configuration and billing ingestion

It allows the user to view all of their cloud charges that went off against their account. On a granular level they can select which charges to monitor or omit and at a time period that they prefer. This way the user can build a custom view of their cloud spending and gain useful operational insights to make informed decisions.



The screenshot shows the CloudSherpa dashboard with a table of AWS resources. The table has columns for Service, Resource ID, and Provider. The resources listed are:

Service	Resource ID	Provider
awskms	*	AWS
AWSCloudFormation	arn:aws:cloudformation:eu-north-1:564907680089:st...	AWS
AWSCloudFormation	arn:aws:cloudformation:eu-north-1:564907680089:st...	AWS
AWSCloudFormation	arn:aws:cloudformation:eu-north-1:564907680089:st...	AWS
AmazonVPC	arn:aws:ec2:eu-north-1:564907680089:network-inter...	AWS
AmazonVPC	arn:aws:ec2:eu-north-1:564907680089:network-inter...	AWS
AmazonVPC	arn:aws:ec2:eu-north-1:564907680089:network-inter...	AWS
AmazonECR	arn:aws:ecr:eu-north-1:564907680089:repository/clo...	AWS
AWSDataTransfer	arn:aws:ecs:eu-north-1:564907680089:task/cloudshe...	AWS
AmazonECS	arn:aws:ecs:eu-north-1:564907680089:task/cloudshe...	AWS

Below the table, there is a pagination bar showing "0 of 35 row(s) selected", "Rows per page: 10", and "Page 1 of 4". To the right of the table, there is a "Time Period" widget set to "30 days".

1. The user can change the name of the KPI widget.
2. They can select the specific billing records they would like to monitor.
3. Users can also select which time window they prefer for the billing.

3.6 Feature 6: Optimization (Recommendations)

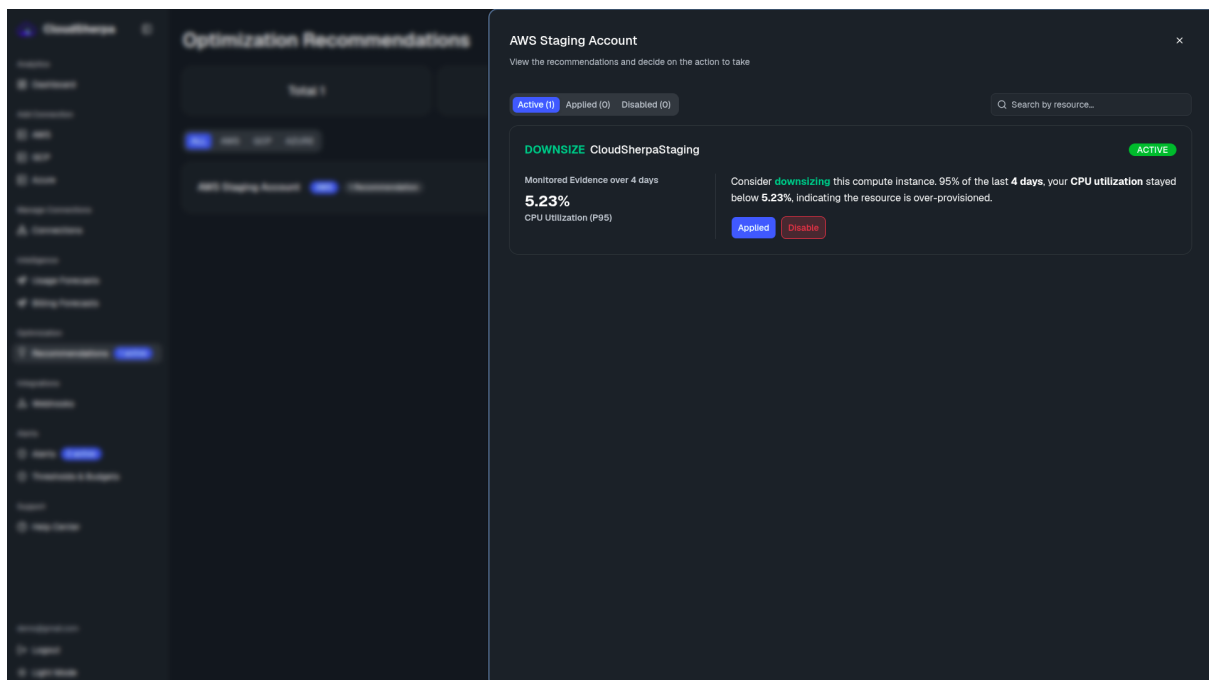
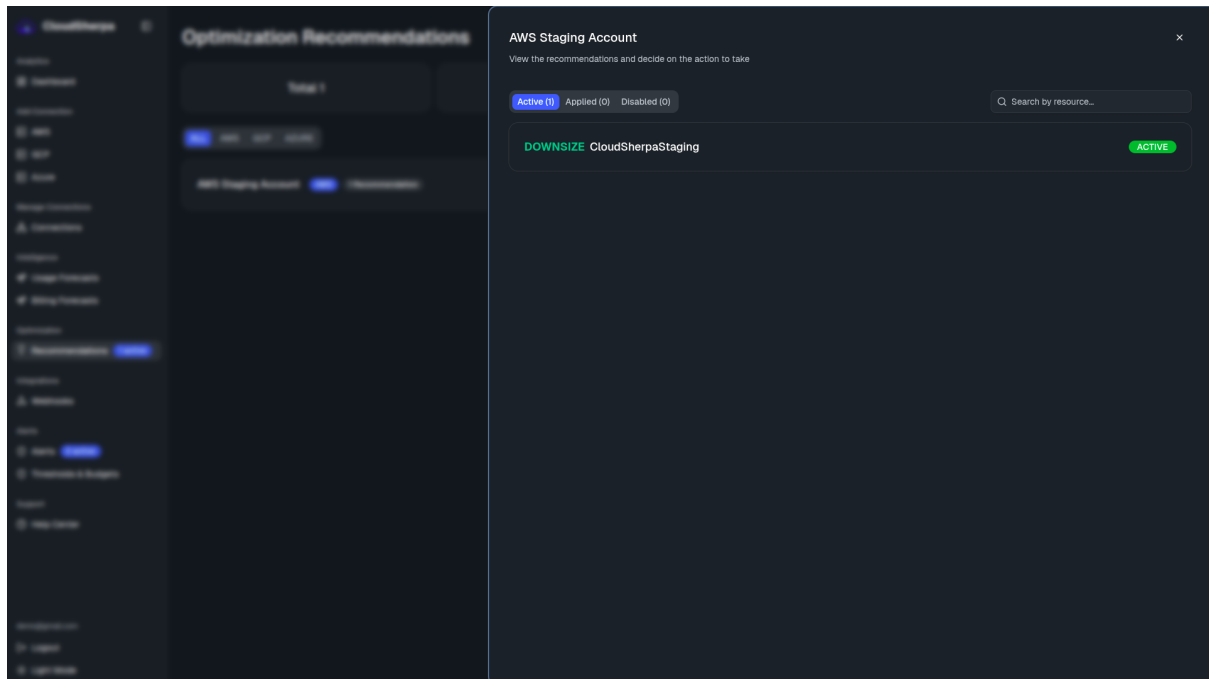
CloudSherpa provides recommendations based on the connections that exist. You can choose whether you would like to accept the recommendation or not.

The screenshot shows the CloudSherpa dashboard with the "Optimization Recommendations" section. It displays a summary of recommendations:

- Total 3
- Active 2
- Applied 0
- Dismissed 1

Below the summary, there are tabs for "ALL", "AWS", "GCP", and "AZURE". A search bar is present with the text "Search by Account...". The recommendations are listed as follows:

- CloudSherpa AWS (AWS) 2 Recommendations
- GCP South Africa Test Singleton (GCP) 1 Recommendation



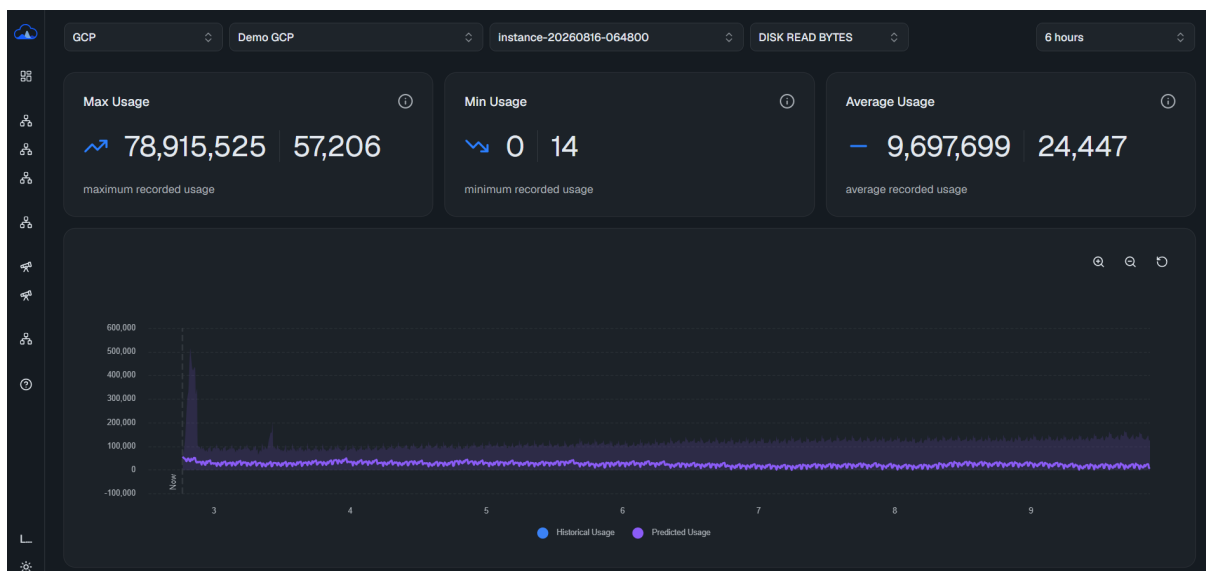
How to navigate to and around the recommendations page:

1. Navigate to the optimizations page using the side navigation bar.
2. A user can click on a connection that is displayed in a list format.
3. A list of recommendations for each resources in that connection will appear.

4. By clicking on a recommendation the user can obtain more information and view the evidence regarding the creation of the recommendation.
5. Click on "Apply" when the user would like to accept that recommendation.
6. Alternatively, the user can click on "Dismiss" to cancel all future recommendation creations of that type for the specific resource.

3.7 Feature 7: Forecasting

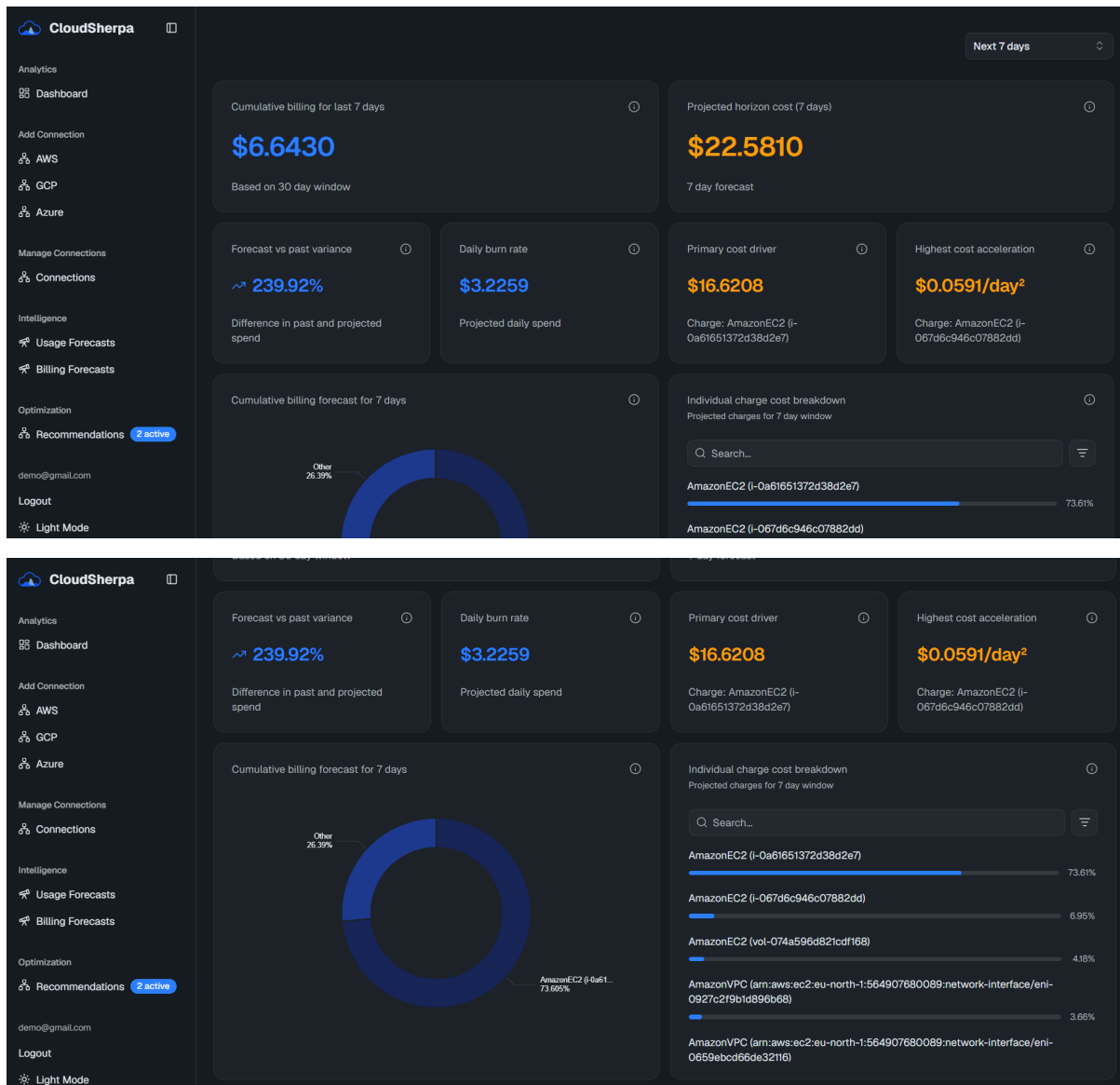
a) Usage forecasting



How to navigate to and around the Usage Forecasting page:

1. Navigate to the Usage Forecasting page using the side navigation bar.
2. Use the toolbar to select the metric of which resource you would like to view the forecast for.
3. The time period selector importantly only controls historical metrics shown.
4. The cards display the statistics.
5. The shaded area indicates the confidence band and the purple line indicates the forecasted value.

b) Billing forecasting



How to navigate to and around the Billing Forecasting page:

1. Navigate to the Billing Forecasting page using the side navigation bar.
2. The user needs to ensure that they have configured billing during setup (this is shown previously in step 2 of the setup wizards).
3. Navigate to billing forecasting page using the side navigation bar.
4. The time period selector, which appears on the top right-hand side corner can be used to control the days that are being forecasted.

3.8 Feature 8: Alerting

CloudSherpa's alerting engine is an active monitoring system that automatically notifies the user when something goes wrong or gets expensive. Instead of watching your dashboards, you can let CloudSherpa watch them for you. Alerts are delivered instantly hence there is no reason to refresh the page for updates on alerts. The alerting engine is made up of: Threshold alerts, which catches metrics that are crossing a limit you defined. Anomaly detection, which catches unusual behaviour and does not require any setup from the user. Billing forecasting alerts, which catches cost that goes beyond your budget set.

a) Anomaly detection

Anomaly detection learns what "normal" looks like for your system and it will flag any behaviour that does not follow that pattern, even if a threshold has not been set for it. It works based on statistical techniques such as a moving average and standard deviation which establishes a baseline of what normal behaviour looks like for each metric. When a resource deviates from that baseline the system detects it and automatically triggers an alert. Anomaly detection flags behaviour that you did not think to check, hence there is no configuration required for it and it is always active.

How to navigate to and around the Alerts:

1. Navigate to the Alerts tabs using the side navigation bar.
2. When clicked upon, the user will get a list of the alerts that relate to anomalies, thresholds and budgets.

3. The user can filter according to anomalies, thresholds or budgets and they can search for a particular alert.
4. The user can opt-in to in app alert notifications, which notifies the user that there has been an alert.
5. More information can be viewed about the alert and an alert can be deleted.
6. Alerts can also be active or disabled by the user.

b) Thresholds

Threshold alerts allow the user to define limits which gives them direct control over what qualifies as a problem for their system. This can be established in the Thresholds & Budgets tab. Once save, CloudSherpa continuously checks incoming data against the threshold and when the metric crosses that line, an alert is pushed to the users screen immediately.

CloudSherpa

Manage Connections
Connections
Intelligence
Usage Forecasts
Billing Forecasts
Optimization
Recommendations (5 active)
Integrations
Webhooks
Alerts
Alerts (11 active)
Thresholds & Budgets
Support
Help Center
demo@gmail.com
Logout
Light Mode

Thresholds & Budgets

Thresholds Budgets

Search thresholds

4 of 4 rules enabled

+ New threshold

ENABLED	RESOURCE	METRIC	CONDITION	SEVERITY
☒	cloudsherpa-test-instance	CPU Utilization	> 5	WARNING
☒	CloudSherpaStaging	CPU Utilization	> 5	WARNING
☒	CloudSherpa Production	CPU Utilization	> 10	CRITICAL
☒	CloudSherpaTestVM	Network In	> 10	WARNING

Showing 1-4 of 4

Page 1 of 1

Rows per page: 10

How to navigate to and around the Thresholds page:

1. Navigate to the Thresholds & Budgets tabs using the side navigation bar.
2. The user will be presented with a list of thresholds that have been set thus far.
3. Relevant information related to that threshold will be displayed in the table.
4. New thresholds can be added via the new threshold button.
5. Thresholds can be edited and deleted.

c) Forecasting alert

Billing forecasting alerts protects users from unexpected cloud bills. If CloudSherpa detects that your costs are increasing suddenly and if it will exceed your budget, the user will receive a warning which gives them time to investigate and fix the problem. For this to work, budgets need to be set under the Thresholds & budgets tab.

ENABLED	SCOPE	BUDGET	WINDOW
☒	Account	USD 1.00	30 days
☒	Cross-account	USD 10.00	30 days
☒	Resource CloudSherpa Production	USD 5.00	30 days

How to navigate to and around the Budgets page:

1. Navigate to the Thresholds & Budgets tabs using the side navigation bar.
2. The user will be presented with a list of budgets that have been set thus far.
3. Relevant information related to that budget will be displayed for the user to view.
4. New budgets can be added via the new budget button.
5. Budgets can be edited and deleted.
6. Budgets can only be added for accounts that have opted into billing when setting up a connection.

3.9 Feature 9: Webhooks

CloudSherpa is built for DevOps and FinOps teams. While the dashboard provides information visually, the webhook extends CloudSherpa to developers as an integration platform. It allows third-party applications to subscribe to events and react to them.

CloudSherpa produces domain events and consumers subscribe to relevant types.

Message queues decouple generation from delivery, preventing slow or unavailable third-party systems from affecting CloudSherps's processing. Requests can be srtpy-tographically signed for verification and users can configure endpoints and selected which event types each subscribes to.

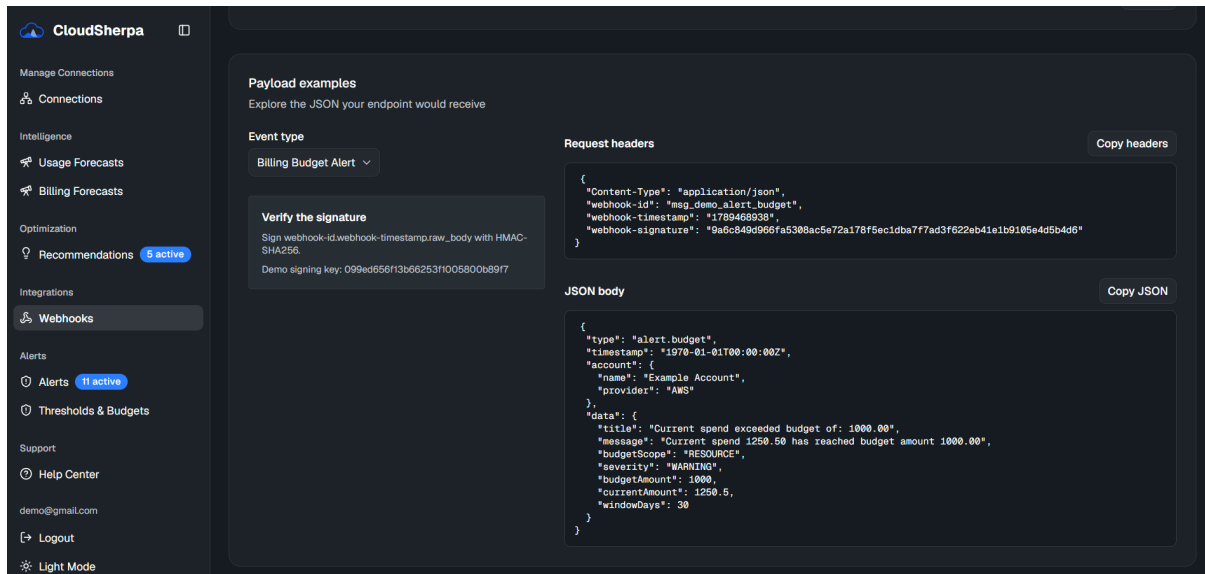
The screenshot shows the 'Webhooks' section of the CloudSherpa dashboard. The left sidebar contains navigation links: Manage Connections, Connections, Intelligence, Usage Forecasts, Billing Forecasts, Optimization, Recommendations (5 active), Integrations, Webhooks, Alerts, Alerts (11 active), Thresholds & Budgets, Support, Help Center, demo@gmail.com, Logout, and Light Mode. The main content area is titled 'Webhooks' with the subtitle 'Manage endpoints and inspect event deliveries'. It features a '+ Add webhook' button in the top right. Below this is a 'Configured webhooks' section with a search bar and a status filter set to 'All statuses'. A table lists the configured webhooks:

Name	Endpoint URL	Events	Accounts	Status	Actions
CloudSherpa Webhook	https://webhooks.gjcs.org/webhooks/cloudsherpa	3 events	All accounts	Active	

Below the table, it says 'Showing 1-1 of 1 webhooks' and 'Page 1 of 1'. The 'Delivery log' section below has a search bar, 'Search' and 'Reset' buttons, and filters for 'All webhooks' and 'All statuses'. A 'Refresh' button is in the top right. The log table shows:

Time	Webhook	Event	Account	Result	HTTP
9/29/2026, 12:24:08 PM	CloudSherpa Webhook	alert.anomaly	gcp-qa	Delivered	202
9/29/2026, 12:24:08 PM	CloudSherpa Webhook	alert.anomaly	gcp-qa	Delivered	202
9/29/2026, 12:24:08 PM	CloudSherpa Webhook	alert.anomaly	gcp-qa	Delivered	202

The 'Add webhook' modal dialog is shown. It has a close button (X) in the top right. The 'Name' field contains 'e.g. Cost reporting'. The 'Endpoint URL' field contains 'https://example.com/webhooks'. There are two main sections: 'Events' and 'Cloud accounts'. The 'Events' section has a search bar and a list with 'Alerts' selected, showing '3 events'. The 'Cloud accounts' section has a dropdown set to 'All connected accounts' and a message: 'All connected accounts. Events from all 4 connected accounts are included. Choose "Specific accounts" to narrow the selection.' At the bottom are 'Cancel' and 'Create webhook' buttons.



How to navigate to and around the Webhooks page:

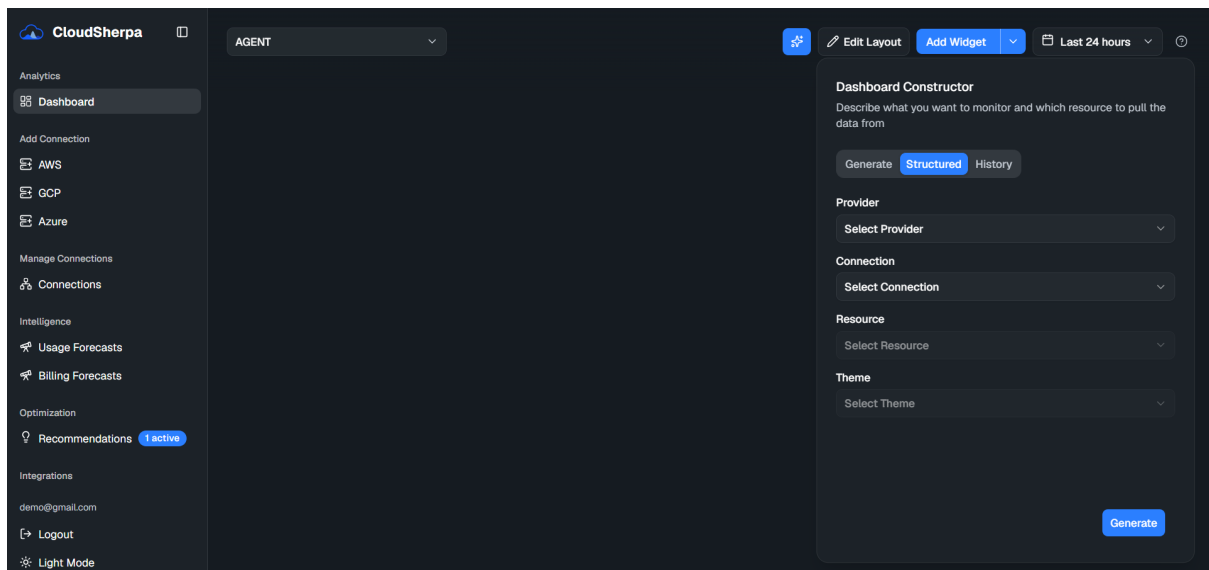
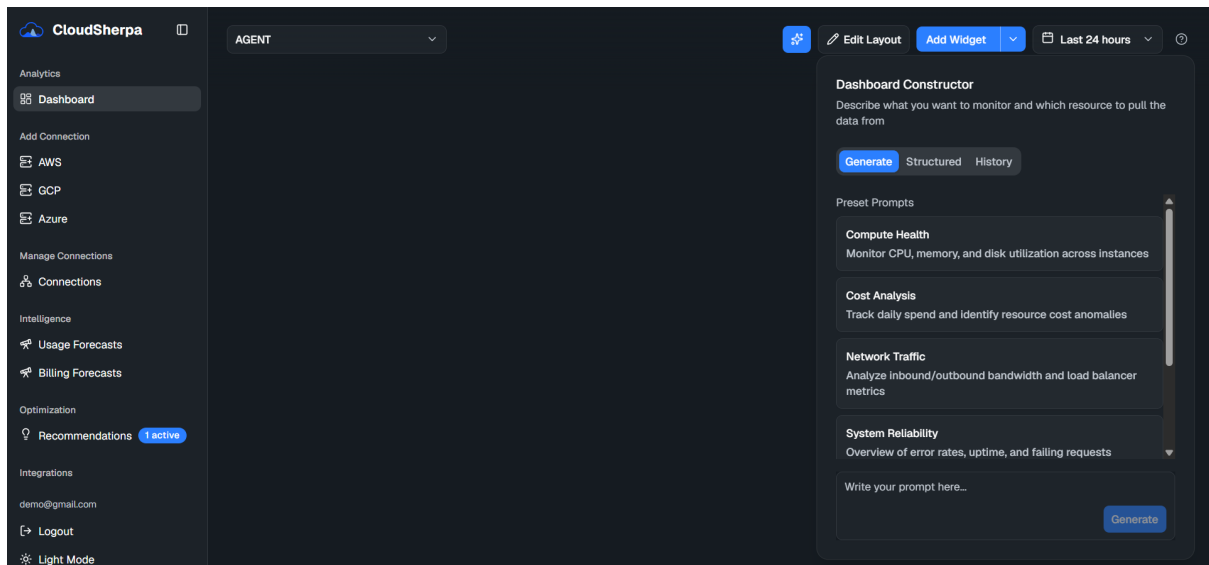
1. Navigate to the Webhooks tabs using the side navigation bar.
2. Users can configure webhooks based on the events and cloud accounts they select.
3. A signing secret will be shown after a webhooks has been configured, which they then need to copy it as it will not be displayed again.
4. The delivery logs provides users information and updates of the webhooks that have already been configures.
5. The payload provides users with how the request headers and JSON body looks.
6. Webhooks can be edited and deleted.

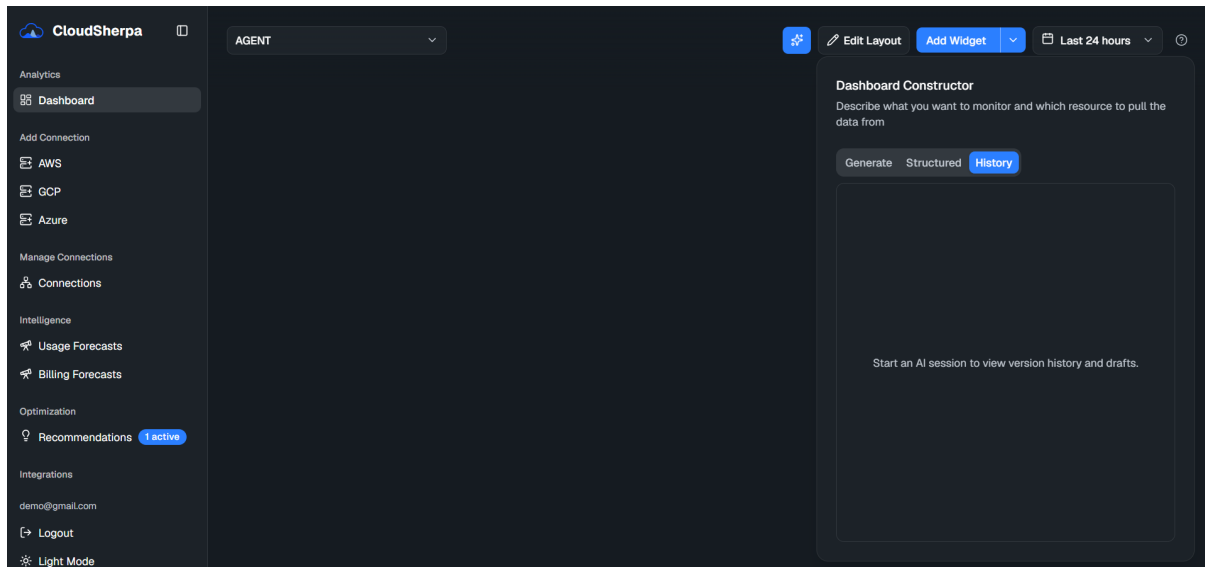
3.10 Feature 10: Agentic dashboard construction

CloudSherpa gives DevOps and FinOps teams a multi-cloud view and Agentic dashboard construction allows users to describe monitoring needs in natural languages. An AI agent automatically builds the dashboard from CloudSherpa's resources, metrics and billing data. This removes the "blank canvas" issues which reduces time-to-value and lets non-technical users generate dashboards without understanding cloud data.

The agent uses a tool-based architecture with an MCP server which exposes controlled tools to retrieve resources, metrics and billing data and to construct or modify dashboard. It keeps it provider neutral and avoids direct cloud API access. Generated configurations need to pass through a validation layer before persistence to ensure

valid resources and that supported metrics are used. Structured prompts replace raw prompts for more deterministic results. The agent supports KPI, line and gauge charts.





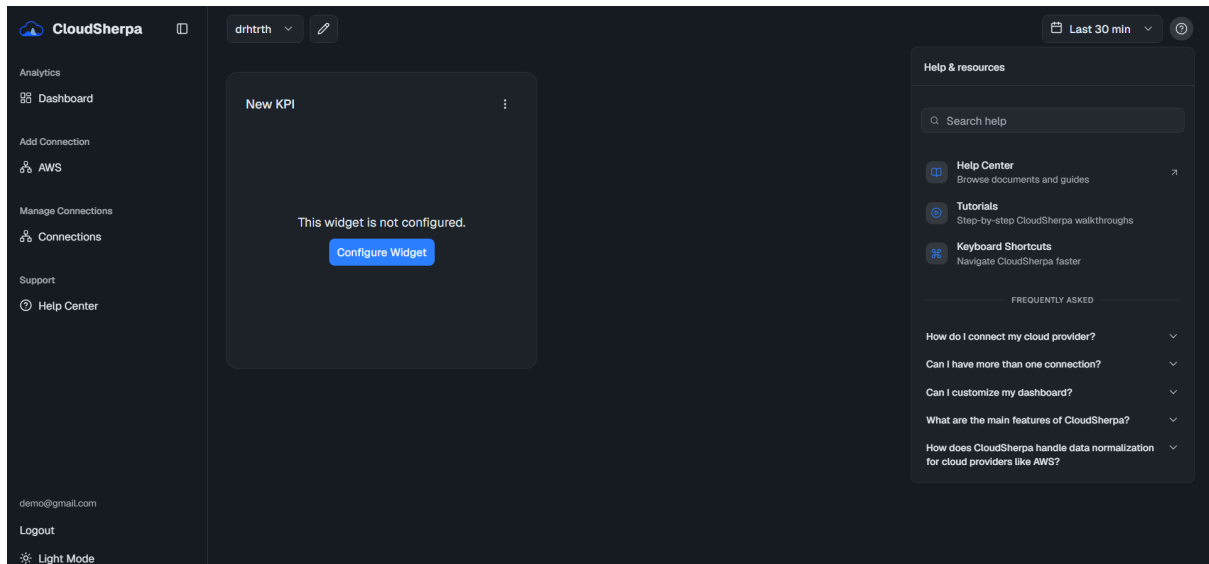
How to navigate to and around the Agentic dashboard constructions:

1. Navigate to the Dashboard tabs using the side navigation bar, if the user is not currently on it.
2. Click on the button that symbolises AI *thestars*.
3. On the generate tab the user has a option if they would like to select a prompt that is listed or they can type in their own prompt.
4. The AI agent will then generate a dashboard according to the prompt, which can be further adjusted by the user before being applied.
5. When navigating to the structured tab, the user can select from the dropdowns as to what the AI agent should include in the dashboard.
6. If the user is not content with the current dashboard, they can navigate to the prompts that were previously inputted and continue from there or further adjust the current dashboard that the AI generated.

4 | In-System Help Menu

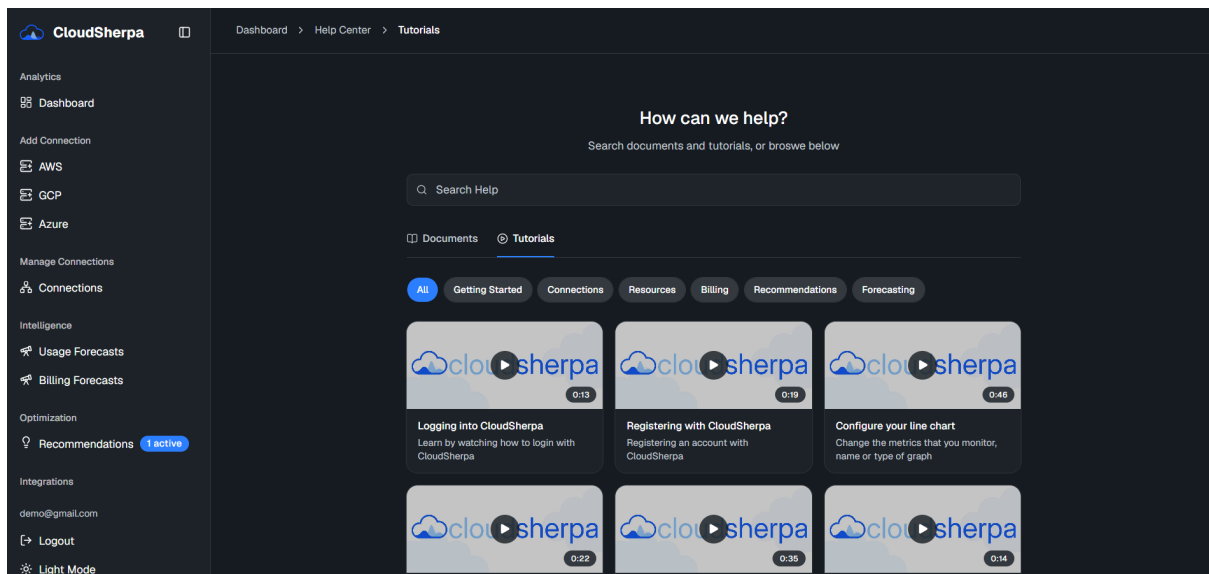
To allow for easy navigation around CloudSherpa, we have included a help menu which users can utilize to familiarize themselves with the features of the platform before delving into it.

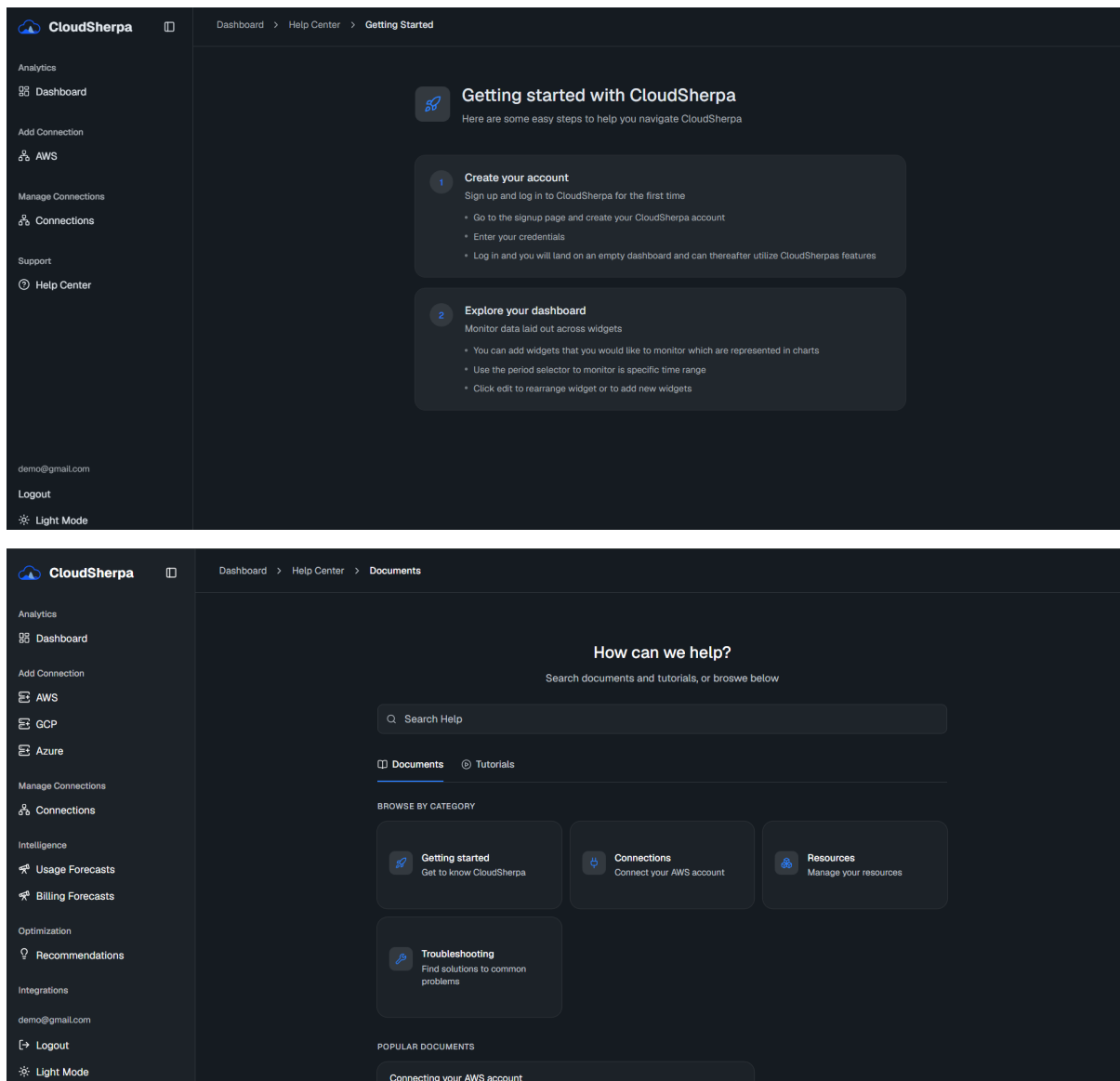
4.1 Accessing Support



The help menu can be accessed from the top-right hand corner of the dashboard by clicking on the ? or in the side navigation bar under support.

When clicked upon, the user is provided with a list of options such as: Help center, tutorials, Keyboard shortcuts and frequently asked questions.





Help center provides detailed documentation and tutorial videos which can help a user to understand how the features of CloudSherpa work. The documents provide step-by-step instructions on how to execute a specific task. It is also accompanied by tutorial videos that demonstrate how the instruction can be executed.

There is detailed documentation on how to setup an AWS, GCP, and Azure connection. Troubleshooting has been provided for users to navigate to when they do not understand how to proceed when following the setup wizard.